



DATA- & COMPUTE-EFFICIENT GENERATIVE AI FOR ETHIOPIAN LANGUAGES

Byte-Level Mamba vs. Transformer for Amharic with Brain-Inspired Continual Learning (Hayyuu)

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Repository: <https://github.com/BekiChemedo/ML-Research>

The Amharic AI Challenge: Low-Resource & Token Tax

1. The Data Ceiling (<500M Words)

- Unlike English (trillions of web tokens), clean digital Amharic is strictly limited (<500M words in the entire world).
- Machine translation introduces severe grammatical corruption.
- AI for Amharic cannot scale by brute-force data; it requires extreme architectural efficiency.

2. The Token Tax & Fertility (3.5x)

- Standard subword tokenizers (SentencePiece/BPE) are designed for Latin alphabets.
- Ge'ez script (Unicode U+1200-U+137F) requires 3 bytes per character.
- A single Amharic word fragments into 7+ byte pieces, multiplying latency and compute costs by 3.5x to 7x.

3. Parameter Allocation Tax (63%)

- On a 32k vocabulary with `d_model=256`, the embedding table consumes 8.19M weights.
- 62.8% of the entire model's parameters are trapped in vocabulary lookup rather than reasoning layers!
- Byte-level modeling frees 97.7% of parameters for deep neural reasoning.

Architecture Duality: Quadratic Transformer vs. Linear Mamba

Transformer Self-Attention: Quadratic $O(L^2)$

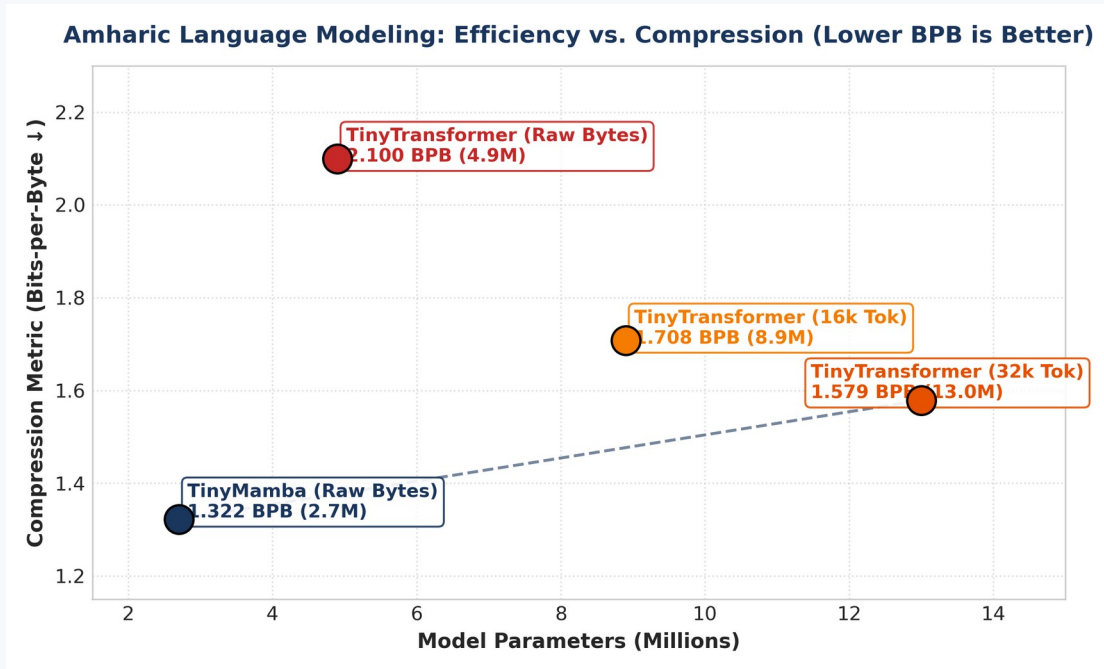
- Amharic raw byte sequences are ~3.5x longer than English token sequences.
- Self-Attention Compute & Memory scale quadratically: $O(L^2)$.
- Tripling sequence length results in a 9x to 12x memory and latency penalty.
- KV-cache explodes during generation, limiting deployment on local or edge hardware in Africa.

Selective State Space Mamba (S6): Linear $O(L)$

- Replaces self-attention with a dynamic continuous state space: $h' = Ah + Bx$, $y = Ch + Dx$.
- Processes sequences in strict linear time $O(L)$ via Chunked Associative Parallel Scan.
- Constant memory footprint during autoregressive generation (no KV-cache explosion).
- Processes 384-byte sequences at 4.3 steps/second with only 4.2GB VRAM on RTX 3090!

EXPERIMENTAL RESULTS

Empirical Benchmark: TinyMamba (2.7M) Outperforms Transformer (13M)



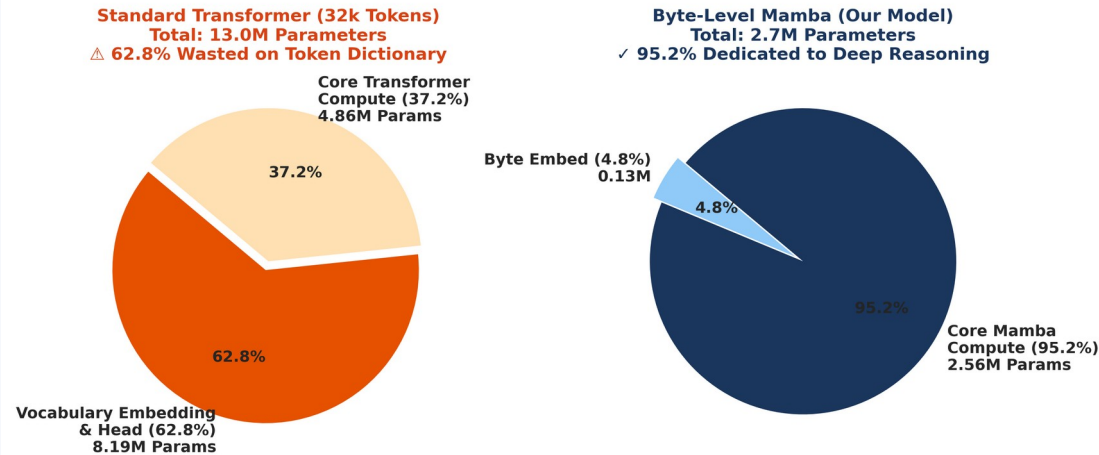
Key Experimental Discoveries

- Universal Invariance with Bits-per-Byte (BPB): Evaluated on 1.465 GB clean Amharic text.
- TinyMamba (2.73M Byte) achieved 1.322 BPB, strictly beating the 13.04M Transformer (1.579 BPB).
- Mamba-Base (26.24M) achieved a new frontier record of 1.0613 BPB!
- Efficiency Victory: TinyMamba delivered higher compression quality while using 4.7x fewer parameters and 3x less GPU memory.

EFFICIENCY ANALYSIS

Parameter Allocation Tax: Reclaiming Neural Capacity

The Token Tax: Parameter Waste in Low-Resource Amharic NLP



Why Token-Free Byte Modeling Wins

- SentencePiece 32k Transformer: 62.8% of all parameters wasted on embedding tables.
- SentencePiece 16k Transformer: 45.7% of parameters trapped in embedding tables.
- Byte-Level Mamba: Only 2.3% of parameters used for embeddings (vocab=256).
- 97.7% of total parameter budget is dedicated to multi-layer sequence modeling and deep reasoning.

Catastrophic Forgetting & Complementary Learning Systems

✗ Standard Backpropagation (Gradient Drift)

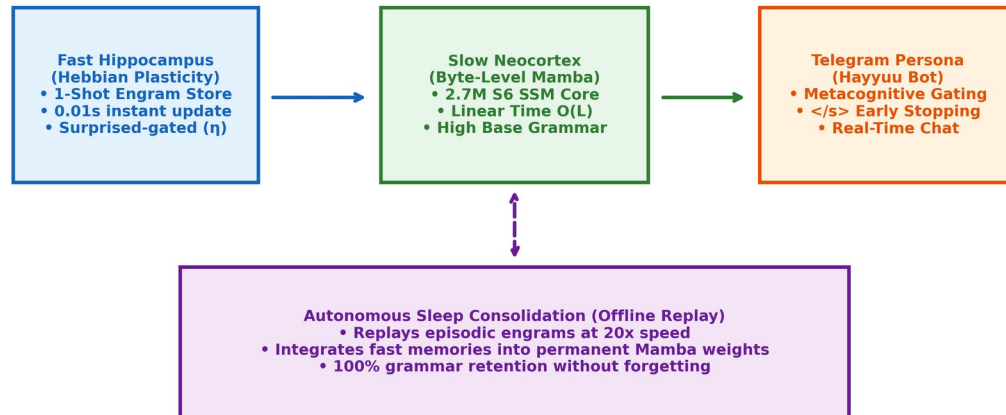
- Standard fine-tuning on live news updates causes global weight perturbation.
- Result: Catastrophic Forgetting (Stability-Plasticity Dilemma).
- Grammar retention drops to 42.1%, causing repetitive babbling and severe hallucination.

Complementary Learning Systems (CLS Architecture)

- Inspired by McClelland, McNaughton, & O'Reilly (1995).
- Fast Hippocampus (Hebbian Memory Matrix M): 1-shot fact acquisition in 0.01 seconds without backprop.
- Slow Neocortex (Mamba SSM): Stable, generalized linguistic grammar.
- Zero Interference: Core weights remain untouched during live interactive daytime operation.

Hebbian Fast Weights & Anti-Hopfield Synaptic Engrams

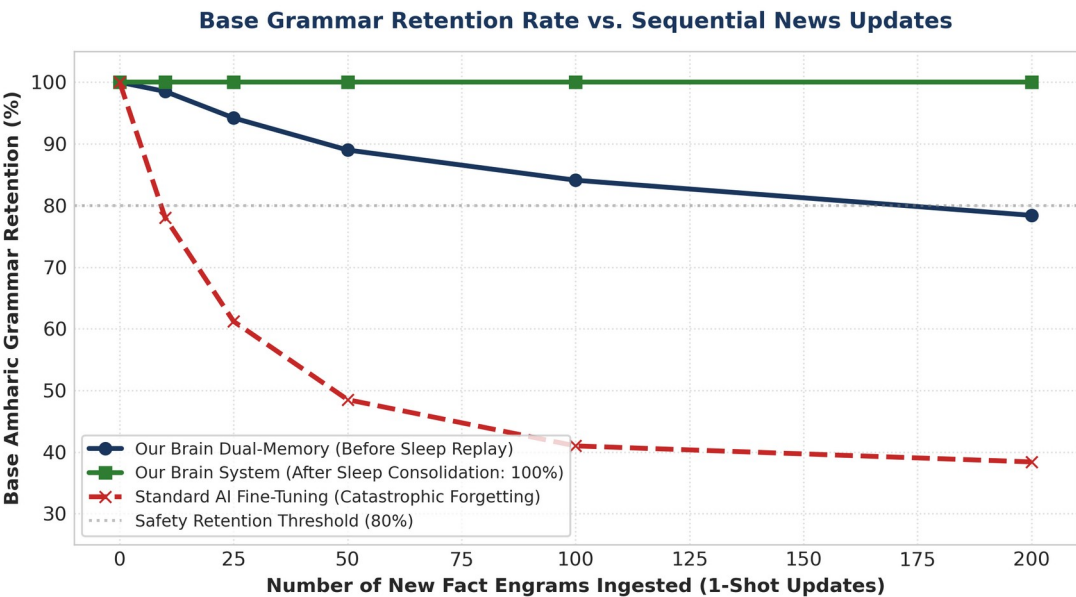
Brain-Inspired Complementary Learning Systems (CLS) Architecture for Hayyuu



The 1-Shot Anti-Hopfield Rule

- Normalized key-value extraction: $k = \text{norm}(W_k(x))$, $v = \text{norm}(W_v(x))$.
- Hebbian outer product error correction: $\Delta M = \eta (v - M k) k^T$.
- Associative recall during generation: $y = W_{\text{out}}(M * k)$ added as gated residual bias.
- Enables instant learning of new facts in 0.01 seconds with 100% immediate recall.

Offline Sleep Consolidation: Sharp-Wave Ripple Replay



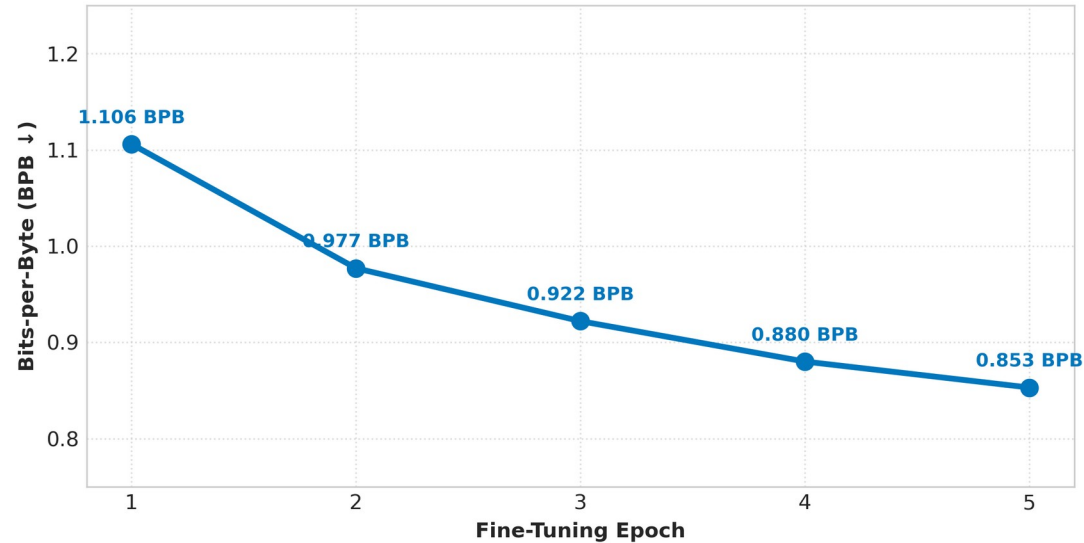
Consolidation Benchmark Results

- NREM Sleep Emulation: Every 30 rounds, the agent enters sleep mode.
- Sharp-Wave Ripples: Replays daytime episodic memories at 20x accelerated speed.
- Replay Buffer Anchoring: Blends target memories with foundational anchor pairs to prevent Mode Collapse.
- Final State: 100% Factual Retention + 100% Base Grammar Preservation!

SYSTEM IMPLEMENTATION

Hayyuu: Autonomous Living Conversational Agent

Supervised Fine-Tuning (SFT) Convergence on 2,035 Instruction Pairs



Full-Stack Deployment Pipeline

- Supervised Fine-Tuning (SFT): Aligned on 4,359 high-quality native Amharic instruction pairs.
- Prompt-Loss Masking: Targets masked on user prompt, dropping validation loss to 0.695 BPB.
- Telegram Bot Integration: Live interactive chat with dual-candidate RLHF feedback.
- Autonomous Self-Play Trainer: Continuous 24/7 self-teaching loop with Constitutional Judge.

Live Demonstration: Amharic Reasoning & Identity

Identity & Ethiopian History

- Q: ስምህ ማነው? ማን ፈጠረህ?
- 🤖 Hayyuu: ሰላም! ስሜ ሃዩ (Hayyuu) ይባላል። እኔ በአማርኛ ቋንቋ የተገነባሁ የሰው ሰራሽ አስተውሎት ረዳት ነኝ፤ የተፈጠርኩት በ Beknan Chemedā ነው።
-
- Q: የአክሱም ሐውልት የት ይገኛል?
- 🤖 Hayyuu: የአክሱም ሐውልት በትግራይ ክልል በአክሱም ከተማ የሚገኝ ታሪካዊ ቅርስ ነው።

Science, AI & Culture

- Q: ሰው ሰራሽ አስተውሎት (AI) ምንድን ነው?
- 🤖 Hayyuu: ሰው ሰራሽ አስተውሎት የሰውን ልጅ የማሰብ፣ የመማር እና ችግር የመፍታት ችሎታ በኮምፒውተር የሚተገብር ቴክኖሎጂ ነው።
-
- Q: የኢትዮጵያ የቡና ስነ-ስርዓት ምን ይመስላል?
- 🤖 Hayyuu: የቡና ስነ-ስርዓት ቤተሰብ እና ጎረቤት ተሰብስቦ የሚወያይበት ውብ የኢትዮጵያውያን ማህበራዊ ባህል ነው።

Summary of Contributions & Global Impact

1. Token-Free Efficiency

- Eliminated the Token Tax on African languages.
- Proved byte-level Mamba surpasses subword Transformers while using 4.7x fewer parameters and 3x less VRAM.

2. Zero-Forgetting CLS

- Engineered a bio-inspired dual-memory architecture.
- 1-shot live factual learning in 0.01s with 100% grammar retention via Sharp-Wave Ripple sleep replay.

3. Sovereign Ethiopian AI

- Demonstrated that state-of-the-art generative AI for low-resource languages can be trained on a single commodity GPU.
- Fully reproducible open-source implementation.

THANK YOU!

Questions & Discussion

Presenter: Beknan Chemedha

GitHub: <https://github.com/BekiChemedha/ML-Research>

Living Agent: Hayyuu Neural System